

Topology First-Year Exam, August 2024, Part 1

For the first five problems, show all your work and support all statements.

1. Let X be a set. Show there is no bijection between X and $\mathcal{P}(X)$, the power set of X .
2. Show that $\mathbb{R}^\omega$ with the box topology is not connected.
3. Show that if X is a compact Hausdorff space, then X is a regular space.
4. Let A be a partially ordered set. The maximum principle states that there is a maximal totally ordered subset B of A . Zorn's lemma states that if every totally ordered subset of A has an upper bound in A , then A has a maximal element. Prove that the maximum principle implies Zorn's lemma.
5. Let (X, d) be a metric space. Suppose there exists some $\varepsilon > 0$ such that for all $x \in X$, the closure $\overline{B_\varepsilon(x)}$ of the ε -ball about x is compact. Prove that X is complete.

Answer the following problems with complete definitions, complete statements, an example, or a short proof.

6. Define what it means for a topological space X to be normal.
7. Let X be compact, and let $C_1, C_2, C_3, \dots$ be a sequence of closed sets in X . If $\bigcap_{i=1}^k C_i$ is nonempty for all $k \geq 1$, then prove that $\bigcap_{i=1}^\infty C_i$ is nonempty.
8. Let $\{A_n\}_{n \in \mathbb{Z}_+}$ be a countable collection of countable sets. Prove that the union $\bigcup_{n \in \mathbb{Z}_+} A_n$ is countable.
9. Show that $[0, 1]$ and $[0, 1)$ are not homeomorphic.
10. Show that $f : \mathbb{R} \rightarrow \mathbb{R}^\omega$ defined by $f(x) = (x, x, x, \dots)$ is not continuous if $\mathbb{R}^\omega$ has the box topology.
11. Give an example of a topological space that is connected but not locally connected.
12. State the Baire Category Theorem.
13. Let (x_n) be a Cauchy sequence in the metric space X . Show that if (x_n) has a subsequence (x_{n_k}) that converges, then (x_n) also converges.
14. Give an example of a metric space (X, d) and a continuous function $f : X \rightarrow X$ satisfying the following:
 - (i) $d(f(x), f(y)) < d(x, y)$ for all distinct $x, y \in X$,
 - (ii) $f(x) \neq x$ for all $x \in X$.
15. Let X be a topological space. Suppose there is a space Y such that the following conditions hold. Prove that X is locally compact.
 - (i) X is a subspace of Y ,
 - (ii) $Y \setminus X$ is a single point, and
 - (iii) Y is compact Hausdorff.